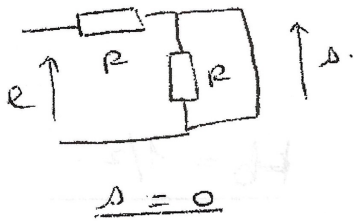
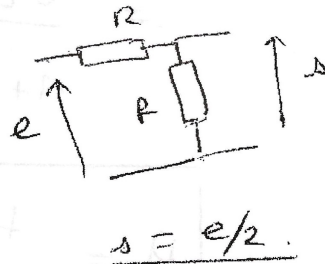


Etude d'un filtre.

1 ABF $j\omega \rightarrow 0$



A HF $j\omega \rightarrow +\infty$

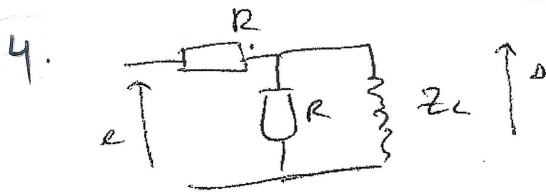


→ passe-haut

2
$$\underline{H} = \frac{s}{e}$$

3
$$G_{dB} = 20 \log G = 20 \log |\underline{H}|$$

$$\varphi = \arg(\underline{H})$$

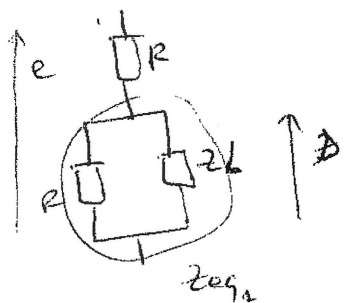


$$s = \frac{Z_{eq1}}{Z_{eq1} + R} e$$

$$\underline{H} = \frac{s}{e} = \frac{1}{1 + \frac{R}{Z_{eq1}}} = \frac{1}{1 + R \left(\frac{1}{R} + \frac{1}{j\omega L} \right)}$$

$$= \frac{1}{1 + 1 + \frac{R}{j\omega L}}$$

$$\underline{H} = \frac{1}{2 + \frac{R}{j\omega L}}$$



$$\frac{1}{Z_{eq1}} = \frac{1}{R} + \frac{1}{j\omega L}$$

$$\underline{H} = \frac{jL\omega}{2jL\omega + 1} = \frac{j\frac{L\omega}{R}}{1 + 2j\frac{L\omega}{R}}$$

$$= \frac{j\frac{\omega}{\omega_c} \times 1/2}{1 + j\frac{\omega}{\omega_c}}$$

$$\omega_c = \frac{R}{2L}$$

$$\boxed{\underline{H} = \frac{H_0 j\omega/\omega_c}{1 + j\frac{\omega}{\omega_c}}}$$

$$\underline{H_0} = 1/2$$

$$5. \quad G = |\underline{H}| = \frac{H_0 \omega/\omega_c}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^2}}$$

$$G_{dB} = 20 \log G$$

A baixa frequência

$$G \sim \frac{H_0 \omega/\omega_c}{1}$$

$$G_{dB} = 20 \log H_0 \frac{\omega}{\omega_c}$$

pende + $20 \log \omega$

A alta frequência:

$$G \sim \frac{H_0 \omega/\omega_c}{\omega/\omega_c} \sim \underline{H_0}$$

$$G_{dB} = 20 \log H_0 = 20 \log 1/2$$

$$= -20 \log 2$$

$$= \underline{-6 \text{ dB}}$$

droite porte

6 BF $\omega \ll \omega_c$

$$7. \quad G_{dB}(\omega_c) = G_{dB}(\text{max}) - 3 \text{ dB}$$

$$8 \quad G = \frac{J}{E} \approx \frac{1}{4} \approx 0,25$$

$$\Rightarrow G_{dB} \approx -12 \text{ dB}$$

$$\Rightarrow \omega \approx 5 \cdot 10^2 \text{ rad/s. (précision limitée)}$$

$$\varphi = 2\pi \cdot \frac{DE}{T} = 2\pi \frac{0,75 \text{ cm max}}{4,5 \text{ cm max}} = 1,0 \text{ rad}$$